

Jason Jaeseung Kim

Biomedical Engineering Undergraduate | Medical Imaging and Computational MRI
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RESEARCH FOCUS

Medical imaging, computational MRI, quantitative imaging, 3D image analysis, and efficient deep learning, with emphasis on reproducible implementation and rigorous quantitative evaluation.

EDUCATION

Yonsei University - B.S. Candidate in Biomedical Engineering

Mar 2021 - Expected
Feb 2027

Mirae Campus | Second Major in Electrical and Electronic Engineering, Sinchon Campus

GPA 4.03 / 4.30	4.50 Scale 4.18 / 4.50	Percentile 97.90 / 100	Department Rank 1 / 49
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Selected coursework: Image Processing and Applications; Digital Signal Processing; Signals and Systems; Electromagnetics I; Digital Logic Laboratory; Computer Programming; Engineering Mathematics; Deep Learning Laboratory; Public Health and Medical Big Data Analysis.

RESEARCH EXPERIENCE

Undergraduate Researcher, Advanced Image Processing Techniques Lab, Yonsei University

Dec 2023 - Feb 2026

Advisor: Prof. Joon Yul Choi | Mirae Campus

- Conducted two first-author medical-imaging studies involving 3D MRI, quantitative imaging, and voxel-wise analysis; project-specific details are withheld before submission at the advisor's request.
- Developed and evaluated 3D deep-learning workflows for medical image regression, quantitative evaluation, and robust validation.
- Built reproducible volumetric MRI pipelines for registration, masking, resampling, normalization, quality control, and downstream analysis.
- Investigated variability across MRI sequences, scanners, vendors, and acquisition parameters, including harmonization-oriented analyses.
- Performed SPM12-based voxel-wise group analyses with covariate adjustment and multiple-comparison correction; prepared manuscript text, figures, tables, and result summaries.

Undergraduate Researcher, Bio-Metamaterials Lab, Yonsei University

Jul 2022 - Aug 2022

Advisor: Prof. Seung Ho Choi | Mirae Campus

- Participated in the initial installation and setup of an ellipsometer and analyzed the compositional and optical properties of biomaterials.
- Built a prototype tensile-strength measurement device and implemented a MATLAB-based user interface for measurement verification.

MANUSCRIPTS AND PRESENTATIONS

- Two first-author journal manuscripts in preparation in medical imaging; project titles, methods, results, and figures are withheld before submission.
- Oral Presentation and Outstanding Paper Award (우수논문상), Oral Presentation Session, Korean Society of Medical and Biological Engineering Fall Conference, Nov 2025.
- Poster, NEURO2026, Jul 2026.

SELECTED PUBLIC RESEARCH AND IMPLEMENTATION

Computational MRI Reconstruction and Motion-Correction Reproduction

2026

Independent implementation using public MRI datasets

- Reproduced and analyzed reconstruction and motion-correction pipelines, including image formation, coil-sensitivity estimation, motion simulation, and iterative reconstruction.
- Conducted failure analysis to identify the effects of undersampling, domain shift, calibration limitations, and reconstruction assumptions.
- Documented implementation decisions and reproducibility constraints for subsequent research development.

SELECTED PROJECT AND LEADERSHIP

Project Lead, CareProofRecord - AI-Based Medical Explanation Video Platform

Jun 2025 - Apr 2026

- Planned and developed an AI service concept to reduce information-transfer errors and improve patient understanding during surgical and clinical explanations.
- Designed a video-generation service architecture intended to mitigate hallucination risk and preserve explanatory context over extended interactions.
- Received KRW 1,000,000 in university startup support after review by the university entrepreneurship program.
- Explored potential clinical use cases with regional medical institutions and other prospective users.

TECHNICAL SKILLS

Programming	Python; MATLAB; Bash; Linux
Deep Learning	PyTorch; 3D image-to-image regression; classification; quantitative model evaluation; Grad-CAM
MRI and Neuroimaging	NIfTI; FSL; ANTs; SPM12; registration; masking; resampling; normalization; VBM; voxel-wise analysis
Statistics and Research Methods	Covariate-adjusted GLM; FWE correction; ROI- and voxel-level analysis; reproducible experimental design

SCHOLARSHIPS

- National Science and Engineering Scholarship, Korea Student Aid Foundation - full tuition (Spring 2023, Fall 2023, Spring 2026).
- Hankyungbum Scholarship, Cheongpa-Hankyungbum Scholarship Foundation, Yonsei University - full tuition (Spring 2022, Fall 2022).
- Academic Excellence Scholarship, Yonsei University (Fall 2021).

HONORS AND AWARDS

Research: Outstanding Paper Award (우수논문상), KOSOMBE Fall Conference, Oral Presentation Session, 2025 Academic: Academic Merit Awards, Yonsei University, 2021-2023 Programming: Excellence Award, Wise Coding Life Algorithm Competition, Yonsei University, 2022	Programming: Grand Prize, ASC Programming Contest, 2021 Entrepreneurship: Grand Prize, MEDICI+ Startup Competition, Yonsei University Wonju LINC+ Project Group, 2021 Entrepreneurship: Honorable Mention, WAVE-LAB Startup Competition, 2025; Encouragement Award, Post Corona-19 Startup Challenge, 2022
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